

Clarifications to confirm with the ISRO mentors

Mentors listed in the problem statement: Pranav Kumar Pandey

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Each question lists what Q-Rex currently assumes and which setting changes if the answer differs.

#	Question	Q-Rex assumption	If the answer differs
1	Is a wide-area (whole-screen) view allowed while searching / re-acquiring , or must acquisition use only the 640×480 camera at its configured FOV? A fixed-FOV scan needs ≈ 12.5 s, so "acquisition ≤ 2 s" needs either a wide view or a zoom lens (the statement lists the FOV as user-defined).	default acquisition = overview (an extra 4×-binned wide sensor, only while searching)	choose Zoom lens (camera-only: the same camera starts at the full-screen FOV, detects in ≈ 0.03 s and zooms in; beacon centred at the default $4^\circ \times 3^\circ$ FOV in ≤ 1.3 s) or <i>Raster scan</i> (fixed FOV, ≈ 5 – 13 s, K01 fails)
2	"Max standard deviation of noise: 20 pixels " — is this 20 grey levels (0–255 intensity noise) or something else (e.g. positional noise of the beacon)?	20 grey levels	gaussianSigma (single parameter)
3	In Benchmark 2, is the .mp4 the whole 2000×2000 screen or the 640×480 camera image? Which pixel convention does the ground-truth use (centre of first pixel = 0.0 or 0.5)?	any size accepted; ≥ 1000 px = whole screen; OpenCV pixel-index convention (centre of first pixel = 0.0)	convention note in the user manual; a constant 0.5 px offset in the centroid error indicates a convention mismatch
4	Is "tracking error" the distance of the beacon from the image centre (boresight) , or the centroid error against ground truth ? Is camera jitter included in the former?	both reported; boresight error excludes the random per-frame jitter (also reported including it)	both numbers are already in every report

#	Question	Q-Rex assumption	If the answer differs
5	Does "acquisition time" start at program start, or when the target first enters the camera FOV?	from program start	acquisitionTimeSec vs timeToCentreSec are both logged
6	What beacon intensity / background level should be assumed (contrast against the background)?	beacon 230, background 20 (grey levels)	targetIntensity, background
7	Is a trained neural network expected for "AI-based", or is adaptive estimation acceptable?	both implemented: classical detector + IMM Kalman filter, plus an optional 74 k-parameter CNN verifier/refiner (trained on simulator data)	toggle <i>AI verifier (CNN)</i> in the Tracker section
8	Are the combined worst cases (jitter ± 20 px/frame and platform ± 20 px/frame and all noises simultaneously) part of the benchmark, or is each disturbance evaluated at its maximum separately?	evaluated separately (all pass); combined case reported honestly (10.8 px mean error at the default $60^\circ/\text{s}^2$ gimbal acceleration (14.3 px at $20^\circ/\text{s}^2$); an ideal causal filter gives ≈ 10.4 px)	—
9	Gimbal acceleration limit — only pan/tilt speed (5–10 $^\circ/\text{s}$) is specified.	$60^\circ/\text{s}^2$ (maxAccel); results are also reported at 20 and $100^\circ/\text{s}^2$ (the combined-disturbance error is 14.3 / 10.8 / 10.6 px)	maxAccel
10	Will the Benchmark-1 scenarios be supplied as configuration files, and in what format? Q-Rex can load a scenario from its JSON config.	scenario JSON / GUI	contact for the format

Draft e-mail

Subject: SIH 26169 — clarifications for the coarse-alignment virtual camera tracker

Dear Sir,

We are working on problem statement 26169 (AI-based virtual camera tracking for FSOC coarse alignment). We would be grateful for clarification on the following points, because they change how we measure and report:

1. Is a wide-area (full-screen) view allowed during search and re-acquisition? With the camera at a fixed $4^\circ \times 3^\circ$ FOV and a $5^\circ/\text{s}$ slew limit, a full-screen scan takes about 12 s, so "acquisition ≤ 2 s" cannot be met by scanning. We implemented both an extra wide-view sensor and a camera-only zoom-lens mode (the FOV is user-defined in the table) — which interpretation do you intend?
2. Is the "maximum noise standard deviation of 20 pixels" meant as 20 grey levels of intensity noise?
3. In Benchmark 2, is the supplied .mp4 the whole 2000×2000 screen, and which pixel-coordinate convention does the ground truth use?
4. Is "tracking error" measured against the image centre (boresight) or against the ground-truth centroid, and should camera jitter be included?
5. Does acquisition time start at program start or when the target enters the FOV?

Thank you, Team Q-Rex